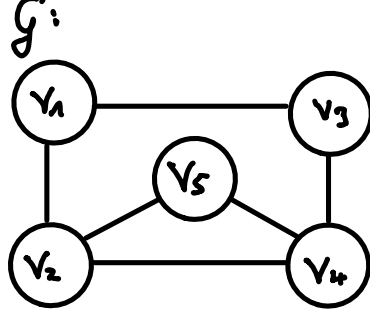
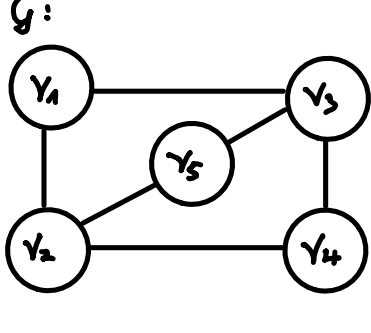


Sheet 3

1. a)



$$\begin{aligned} (1, \{1, 1, 1\}) &\rightarrow 1 & (1, \{1, 1, 1, 1\}) &\rightarrow 2 & (1, \{1, 2\}) &\rightarrow 3 \\ (1, \{2, 2\}) &\rightarrow 4 & (2, \{1, 1, 1, 1\}) &\rightarrow 5 & (2, \{1, 1, 2\}) &\rightarrow 6 \end{aligned}$$

The label distributions diverge after 2 iterations.

(see below)

Since this test is a necessary condition for isomorphism:

$$G \neq G'$$

This result is expected because:

$$(v_2, v_4) \in E(G') \wedge \deg(v_2) = \deg(v_4) = 3$$

and no such edge exists in G .

Test

$$\{4, 4, 4, 5, 5\} \neq \{3, 3, 4, 6, 6\}$$

$t=3$

i	$h_{v_i}^t$	$h_{v_i}^{t*}$
1	4	3
2	5	6
3	5	3
4	4	6
5	4	4

$t=2$

i	$h_{v_i}^t$	$M_{v_i}^t$	$h_{v_i}^{t*}$	$M_{v_i}^{t*}$
1	1	$\{2, 2\}$	1	$\{1, 2\}$
2	2	$\{1, 1, 1\}$	2	$\{1, 1, 2\}$
3	2	$\{1, 1, 1\}$	1	$\{1, 2\}$
4	1	$\{2, 2\}$	2	$\{2, 1, 1\}$
5	1	$\{2, 2\}$	1	$\{2, 2\}$

$t=1$

i	$h_{v_i}^t$	$M_{v_i}^t$	$h_{v_i}^{t*}$	$M_{v_i}^{t*}$
1	1	$\{1, 1\}$	1	$\{1, 1\}$
2	1	$\{1, 1, 1\}$	1	$\{1, 1, 1\}$
3	1	$\{1, 1, 1\}$	1	$\{1, 1\}$
4	1	$\{1, 1\}$	1	$\{1, 1, 1\}$
5	1	$\{1, 1\}$	1	$\{1, 1\}$

2. a)

A

	1	2	3	4	5
1	0	1	0	0	0
2	1	0	1	0	0
3	0	1	0	1	0
4	0	0	1	0	1
5	0	0	0	1	0

$L \approx$

	.5	.40	0	0	0
	.40	.33	.33	0	0
	0	.33	.33	.33	0
	0	0	.33	.33	.40
	0	0	0	.40	.5

2. b)

$$x \in \mathbb{R}^{5 \times 1}$$

Exists a $W \in \mathbb{R}^{3 \times 1}$ and $i \in \{-1, 0, 1\}$ with:

$$(W * x)_i = \sum_{\tau=-1}^1 W_{\tau} x_{i+\tau} = W_{-1} x_{i-1} + W_0 x_i + W_1 x_{i+1}$$

We assume zero-centered indexing for W and x .

$$x = [x_{-1} \ x_0 \ x_1]^T \quad W = [W_{-1} \ W_0 \ W_1]^T$$

Let assume there is a W s.t. $\forall i \in \{-1, 0, 1\}. (W * x)_i = (g_0 * x)_i$

$$\sum_{\tau=-1}^1 W_{\tau} x_{i+\tau} = (Lx)_i = (Lx)_i$$

$$\sum_{\tau=-1}^1 W_{\tau} x_{i+\tau} = \sum_{j=-2}^2 L_{i,j} x_j$$

$$\begin{aligned} W_{-1} x_{i-1} + W_0 x_i + W_1 x_{i+1} &= L_{i,-2} x_{-2} + L_{i,-1} x_{-1} + L_{i,0} x_0 + L_{i,1} x_1 + L_{i,2} x_2 \\ &= L' x \end{aligned}$$

$$\text{s.t.} \quad L' = \begin{bmatrix} .40 & .33 & .33 & 0 & 0 \\ 0 & .33 & .33 & .33 & 0 \\ 0 & 0 & .33 & .33 & .40 \end{bmatrix}$$

For $i=-1$:

$$L_{(-1),(-2)} \approx .40 \Rightarrow W_{-1} \approx .40$$

For $i=0$:

$$L_{0,(-1)} \approx .33 \Rightarrow W_{-1} \approx .33 \quad \checkmark$$

$\Rightarrow$ No such W can exist since $L_{(-1),(-2)} \neq L_{0,(-1)}$.

2. c)

A

L

	1	2	3	4	5
1	0	1	0	0	1
2	1	0	1	0	0
3	0	1	0	1	0
4	0	0	1	0	1
5	1	0	0	1	0

	.33	.33	0	0	0
	.33	.33	.33	0	0
	0	.33	.33	.33	0
	0	0	.33	.33	.33
	0	0	0	.33	.33

$$W = [.33 \ .33 \ .33]^T$$